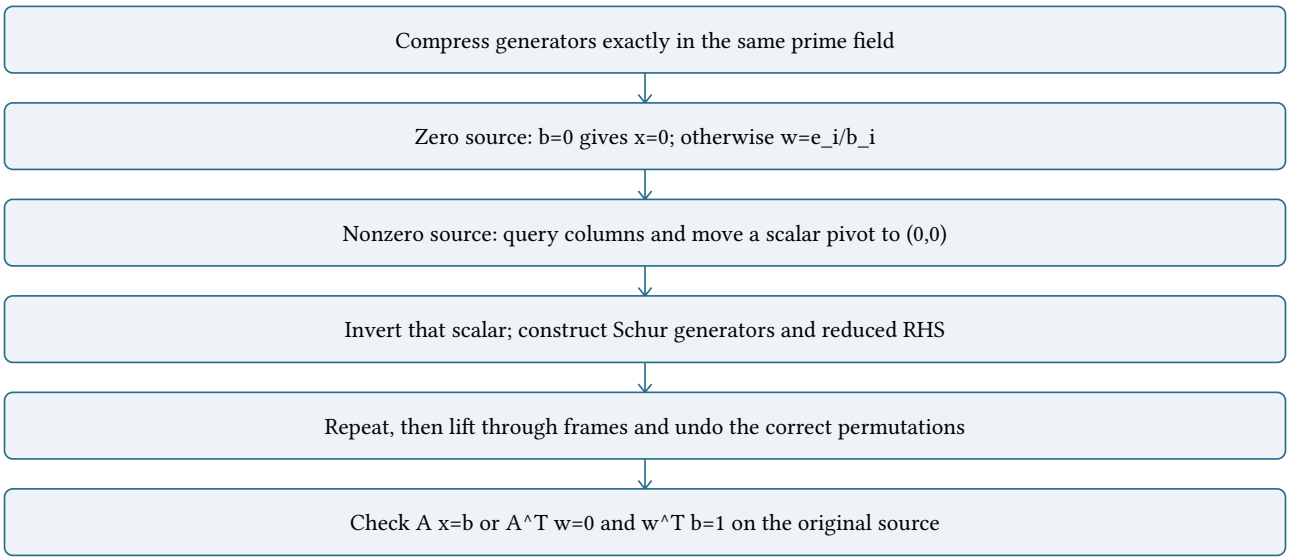
Structured solve or normalized witness

2026-11-02 · Derived implementation; syntax checked · Zero numerical trials

Construction. For a square companion displacement $Z_a A - A Z_b = GK^T$ over a caller-certified prime field, with $a \neq b$, combine exact compression, deterministic scalar pivots and Schur lifts. No supplied matrix solve is used: every leading solve multiplies by an explicitly inverted nonzero scalar.



The zero-source branch terminates at its level and proceeds to unwinding; only the nonzero branch continues through pivoting and Schur reduction.

Column transport. For $u = e_i - e_j$ and $Q = I - uu^T$, represent AQ by

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Gc = [G  | A u | -A Z_b u]
Kc = [QK | Z_b^T u | u]
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Two forward queries add two generator columns. Existing row transport adds two transpose queries when nontrivial. For $B = PAQ$, solutions map as $x = Qy$; left-null witnesses map as $w = P^T v$. RHS maps as Pb. Confusing these two permutation paths would invalidate certificates.

Proof. Distinct wraps make the square Sylvester operator invertible, so zero displacement means $A = 0$. A nonzero matrix has a nonzero queried entry. After transport, that entry is an invertible scalar E. Block elimination gives $\operatorname{rank}(A) = 1 + \operatorname{rank}(S)$, and Schur residual and pairing identities transport either answer. Each step reduces dimension by one. Control flow uses an explicit frame stack, not recursive Python calls. The final root equation is checked with a charged original-source application.

Charged schedule. A node searches $j+1$ columns through its first nonzero column j. With compressed width t and nontrivial column/row swaps c, p, transported width $w = t + 2c + 2p$. Schur setup uses $2w+6$ source queries and $2w+4$ scalar solves. RHS reduction and unwinding each add one query and one scalar solve. Compression setup and one inversion per pivot are additional. The scalar terminal omits Schur work. Root witness verification also charges a dot product. All field arithmetic uses the existing Meter.

Scope and next checks. Singular compatible systems may return zero on free coordinates. An inconsistent system yields $A^T w = 0$ and $w^T b = 1$; feature annihilation $F^T w = 0$ remains separate and explicitly unchecked. Derived rank is not independently certified at runtime. Frames and compression traces can require substantial aggregate storage; scalar depth and width growth do not inherit balanced-inverse bounds. No speed measurement is claimed.

Next freeze full pivot, singular-system, both-transport-direction, corruption, rank-reference and exact-accounting checks, preserving upstream cohorts. Numerical execution remains behind the recorded cost gate.

Sources. solve.py and DERIVATION.md; local pivot, compression, recursive and Schur-lift primitives. Bostan et al. (2017), Section 6, printed pp. 764–765, DOI 10.1137/16M1062855: established structured rank-profile/solve context. Its randomized preconditioning and fast complexity are not assumed here.